

KOTKA HAAPASAARI, Finland

WMO: 029670

Lat: 60.287N

Lon: 27.185E

Elev: 18

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.1	3.7	-5.6	4.1	-0.6	-0.6	5.3	4.0	36.4	28.1	33.7	28.6	10.9	20	0.779

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	76.1	68.6	73.6	67.0	71.3	65.3	70.0	74.9	68.1	73.2	66.0	70.6	9.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.0	102.9	73.8	66.0	95.8	72.1	63.8	88.7	69.4	34.0	74.8	32.4	73.3	30.7	70.7	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.9	30.4	27.6	-1.5	79.2	9.3	5.0	-8.2	82.8	-13.7	85.7	-18.9	88.5	-25.7	92.2		
(5)			-2.0	70.3	9.2	2.5	-8.6	72.1	-14.0	73.6	-19.2	75.0	-25.9	76.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.7	25.5	23.2	28.0	37.6	49.0	57.2	64.8	63.8	56.4	46.1	38.3	32.8		
	DBStd	15.61	9.49	10.21	7.64	4.48	6.35	5.00	5.29	4.39	4.57	5.50	6.19	7.04		
	HDD50	3700	760	749	683	373	96	4	0	0	8	142	351	534		
	HDD65	7886	1225	1169	1148	822	495	240	70	74	259	584	801	999		
	CDD50	1397	0	0	0	1	66	220	459	429	199	23	0	0		
	CDD65	108	0	0	0	0	0	6	64	38	0	0	0	0		
	CDH74	187	0	0	0	0	0	10	143	34	0	0	0	0		
	CDH80	9	0	0	0	0	0	1	8	0	0	0	0	0		
	WSAvg	14.5	16.2	14.1	14.1	12.5	12.0	12.7	11.7	12.8	14.7	16.3	17.9	18.7		
	PrecAvg	26.2	2.1	1.7	1.5	1.3	1.6	2.2	2.5	2.9	2.5	2.8	2.7	2.3		
	PrecMax	36.0	4.0	4.3	3.1	2.8	4.5	4.9	4.4	6.1	5.6	6.5	6.1	5.8		
	PrecMin	19.4	0.2	0.1	0.0	0.2	0.4	0.4	0.2	0.3	0.6	0.7	0.3	0.6		
	PrecStd	3.9	0.9	1.0	0.7	0.6	1.1	0.9	1.2	1.4	1.5	1.5	1.2	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	37.8	41.3	55.4	69.5	74.7	80.5	77.4	67.2	57.6	49.9	45.8		
		MCWB	39.2	35.4	36.7	47.3	57.6	64.1	70.0	69.9	62.1	55.0	48.3	44.4		
	2%	DB	39.2	36.0	38.7	48.8	65.3	70.5	77.6	74.4	64.8	56.3	48.5	43.8		
		MCWB	38.1	34.7	35.4	42.9	55.7	62.5	69.5	67.8	60.6	54.1	47.1	42.3		
	5%	DB	37.8	35.0	37.2	45.8	61.9	66.7	75.2	72.4	63.3	55.2	47.2	42.3		
		MCWB	36.7	34.0	34.9	40.9	54.8	59.7	68.3	66.2	59.5	53.1	45.8	40.9		
	10%	DB	36.3	34.0	35.9	43.4	58.0	64.1	72.8	70.2	62.1	53.7	45.7	40.8		
		MCWB	35.3	33.2	34.2	39.6	52.4	58.3	66.8	64.5	58.6	51.4	44.3	39.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.7	36.1	37.9	47.7	60.1	66.9	72.6	71.1	63.7	56.0	48.6	44.8		
		MCDB	40.2	36.7	40.0	54.5	66.3	73.2	77.6	76.1	66.1	56.9	49.3	45.6		
	2%	WB	38.4	35.2	36.2	44.1	57.4	63.4	70.9	69.1	62.0	54.7	47.4	42.7		
		MCDB	39.0	35.8	37.7	48.2	63.3	68.6	75.8	74.0	64.1	55.9	48.3	43.6		
	5%	WB	36.9	34.2	35.3	41.7	55.3	61.2	69.3	67.3	60.6	53.4	46.0	41.1		
		MCDB	37.8	34.9	36.7	44.8	60.8	65.9	74.2	71.7	62.9	54.9	47.0	42.2		
	10%	WB	35.5	33.3	34.5	40.1	52.9	59.0	67.5	65.3	59.3	51.9	44.5	39.6		
		MCDB	36.3	33.9	35.6	42.8	58.1	63.3	72.7	69.3	61.8	53.6	45.5	40.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	7.1	8.5	7.4	8.6	7.6	7.2	6.1	5.4	5.0	4.7	5.4		
		MCDBR	4.8	4.5	6.6	10.6	13.5	11.4	9.5	7.7	5.9	4.7	4.8	4.8		
		MCWBR	5.1	4.2	4.8	6.1	7.6	6.2	5.0	4.8	4.5	4.7	5.1	4.9		
	5% WB	MCDBR	4.9	4.3	5.8	9.3	12.3	10.1	8.7	6.5	5.1	4.6	4.9	4.9		
		MCWBR	5.2	4.1	4.5	5.9	7.8	6.3	5.1	4.7	4.5	4.8	5.4	5.1		
(40) Clear-Sky Solar Irradiance	taub		0.279	0.311	0.333	0.371	0.369	0.366	0.388	0.381	0.354	0.335	0.307	0.386		
	taud		2.137	2.156	2.199	2.274	2.405	2.462	2.419	2.435	2.490	2.386	2.208	2.602		
	Ebn at Noon		148	211	248	261	272	275	264	257	245	206	137	104		
	Edh at Noon		14	23	31	35	33	32	32	30	24	19	13	9		
(44) All-Sky Solar Radiation	RadAvg		79	272	743	1226	1683	1752	1748	1334	795	329	88	43		
	RadStd		17	57	94	133	181	120	157	137	94	48	15	6		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (47 neighbors)	+0.88	N/A	N/A	N/A	N/A	-302	-319	N/A	N/A	

Nomenclature: See separate page